

CastFSR: A Fast–Slow–Reflect Agentic Reasoning Framework for Context-Aware Time Series Forecasting

Xiaoyu Tao, Mingyue Cheng, Bokai Pan, Chuang Jiang, Huanjian Zhang,
Tian Gao, Yaguo Liu, Qi Liu, and Enhong Chen

State Key Laboratory of Cognitive Intelligence, University of Science and Technology of China
Hefei, Anhui Province, China

{mycheng, qiliuql, chenh}@ustc.edu.cn,

{txytiny, bk2585934928, jiangchuang, zhjustc, ustc25gt, ygliu}@mail.ustc.edu.cn

Abstract

Time series forecasting is fundamental to decision-making in complex systems, where future dynamics are shaped not only by historical observations but also by evolving contextual features. Recent advances in large language models (LLMs) have extended forecasting beyond numerical pattern extrapolation toward semantic integration and context-aware reasoning. However, existing approaches still struggle to effectively couple intrinsic temporal dynamics with complex contextual features, often lacking explicit mechanisms to validate forecasts against temporal and domain constraints. Consequently, they cannot reliably determine which contexts matter, reason about how they reshape future dynamics, or produce knowledge-consistent forecasts. In this work, we propose CastFSR, an agentic framework that formulates context-aware time-series forecasting as a Fast–Slow–Reflect workflow. During fast-thinking forecasting, CastFSR profiles the observation and autonomously selects suitable lightweight forecasters to construct a data-driven forecast prior. During slow deliberative reasoning, it retrieves relevant evidence from long-range contextual histories, adaptively identifies context-specific look-back windows, and reasons about how contextual features alter future dynamics to refine the forecast prior. During reflective evaluation, CastFSR iteratively refines the forecast to enforce temporal, contextual, and domain consistency. CastFSR can be directly instantiated with off-the-shelf general-purpose LLMs, enabling training-free inference without task-specific adaptation. Alternatively, we introduce a two-stage post-training strategy comprising supervised fine-tuning (SFT) and multi-turn reinforcement learning (RL), which enables compact LLMs to internalize CastFSR’s orchestration and reasoning capabilities for efficient inference with post-trained models. Extensive experiments across public datasets demonstrate that CastFSR achieves superior performance against representative baselines.¹

Introduction

Time series forecasting (TSF) plays a critical role in decision-making systems, including energy management (Zhang et al. 2020), financial analysis (Palaskar et al. 2024), transportation systems (Guo et al. 2019), and environmental sensing (Lu et al. 2026). In these domains, a time series is not merely a numeric sequence, but an interface to a system: recorded values

¹Our code is available at <https://github.com/Xiaoyu-Tao/CastFSR>.

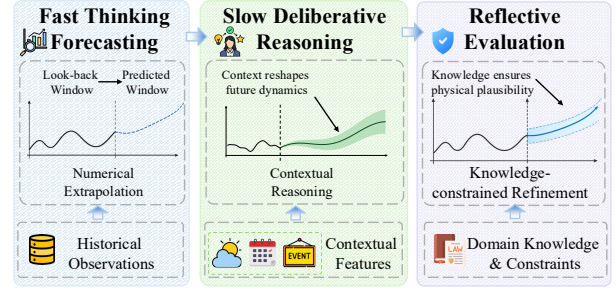


Figure 1: Illustration of the CastFSR, which integrates numerical extrapolation, contextual reasoning, and knowledge-constrained reflection for context-aware TSF.

reflect latent evolution under changing conditions (Holland 1992; Ji et al. 2023), while future trends are shaped by historical observation and contextual features (Tao et al. 2026). These factors cannot always be inferred from historical observations alone, requiring methods to reason over temporal patterns and relevant external contextual features (Ashok et al. 2024; Shao et al. 2024a).

Numerous approaches have been developed for TSF (Guan et al. 2026; Jin et al. 2023), spanning statistical models (Box et al. 2015), machine learning methods (Masini, Medeiros, and Mendes 2023), deep neural networks (Cheng et al. 2025b), and foundation models (Woo et al. 2024). Despite their success in temporal modeling, these approaches mainly rely on historical observations and provide limited support for evolving contexts (Cheng et al. 2025a; Gao et al. 2024). Recent advances in LLMs have opened new opportunities for context-aware forecasting by extending time series modeling toward semantic integration and reasoning. Existing studies have explored representing temporal values in language-compatible forms to leverage pretrained knowledge (Jin et al. 2024). More recent studies have extended LLM-based forecasting toward reasoning-driven context-aware prediction with explicit multi-step deliberation (Cheng et al. 2026; Wang et al. 2025). However, existing approaches still struggle to couple temporal dynamics with contextual features, lacking mechanisms to identify relevant contexts, reason about impacts, and validate forecasts against domain constraints (Jia et al. 2024; Chang et al. 2025).

Based on the above analysis, we formulate context-aware TSF as an agentic sequential decision-making problem that coordinates numerical forecasting, contextual evidence, and reflective validation. As shown in Figure 1, rather than treating LLMs as numerical predictors, we leverage their reasoning capabilities to orchestrate forecasting models, integrate heterogeneous contextual information, and refine forecast priors through evidence-grounded deliberation. However, realizing such a Fast–Slow–Reflect forecasting agent introduces several challenges. First, the agent must construct reliable forecast priors by profiling historical observations and selecting suitable forecasting models, rather than relying on LLM generation (Ji et al. 2023). Second, it must retrieve future-relevant contextual evidence and reason about how contextual factors influence future dynamics, requiring adaptive retrieval and coordination among diverse tools (Wu et al. 2026). Third, it must validate forecasts against temporal regularities, contextual consistency, and domain constraints (Zhang et al. 2025). Finally, applying such a workflow requires flexible deployment strategies that enable the reasoning process to be executed (Deng et al. 2022).

To address the challenges outlined above, we propose CastFSR, an agentic Fast–Slow–Reflect framework that formulates context-aware TSF as a sequential decision-making process. In fast-thinking forecasting, CastFSR profiles historical observations and autonomously selects suitable lightweight forecasters to construct a data-driven forecast prior. In slow deliberative reasoning, it retrieves future-relevant evidence from long-range contextual histories, adaptively identifies context-specific look-back windows, and reasons about how contextual factors reshape future dynamics to refine the forecast prior. In reflective evaluation, CastFSR iteratively evaluates and adjusts candidate forecasts to enforce temporal regularities, contextual consistency, and domain constraints. Importantly, CastFSR supports flexible deployment: it can be directly instantiated with off-the-shelf LLMs for training-free execution, or distilled into compact models through supervised fine-tuning and multi-turn reinforcement learning to internalize the orchestration and reasoning capabilities. Through this design, CastFSR moves beyond conventional pattern extrapolation by enabling forecasting systems to determine when historical patterns are reliable, when contextual evidence should modify predictions, and when further evaluation is required.

Our main contributions can be summarized as follows:

- We introduce a Fast–Slow–Reflect formulation that re-frames context-aware time series forecasting as an agentic process integrating numerical prediction, contextual reasoning, and knowledge-constrained refinement.
- We propose CastFSR, which constructs forecast priors via autonomous forecaster selection, refines them through context retrieval and influence reasoning, and validates predictions with temporal, contextual, and domain consistency checks.
- We instantiate CastFSR with general-purpose LLMs and compact models, both achieving strong performance against representative baselines, with two-stage training improving overall performance on most benchmarks.

Related Work

This section reviews two complementary lines of related work: advances in time series forecasting and developments in agentic decision-making systems.

Time Series Forecasting

Time series forecasting has evolved from statistical models to machine learning, deep learning, and foundation models. Early approaches, including autoregressive models (Winters 1960) and state-space models (Hyndman and Khandakar 2008), provide interpretable formulations for extrapolating historical patterns. Learning-based methods, such as recurrent networks (Wang et al. 2019), temporal convolutions (Cheng et al. 2025a), and Transformers (Shi et al. 2025), improve the modeling of complex temporal dependencies. More recently, foundation models (Ansari et al. 2025) and large-scale pre-trained architectures (Xue and Salim 2023) enhance generalization by learning transferable patterns from large time series corpora. Despite these advances, most methods still infer future values primarily through numerical pattern extrapolation. To move beyond historical observations, many studies incorporate additional information (Jin et al. 2024). Multivariate methods model dependencies among correlated series, while context-aware approaches introduce contextual features (Liu et al. 2025a). However, such information is typically supplied as predefined inputs within a fixed, single-pass inference process (Wu et al. 2021). Consequently, existing methods remain limited in actively retrieving future-relevant context, assessing whether historical patterns remain applicable, and adaptively revising forecasts when contextual factors reshape future dynamics.

LLM-driven Agent

Research on agentic and decision-making systems has a long history in artificial intelligence, particularly in reinforcement learning and sequential decision optimization (Luo et al. 2025). Classical reinforcement learning learns policies that map states to actions in dynamic environments and has been widely applied to control and decision-making problems. These approaches provide a principled framework for sequential decisions, long-term objectives, and delayed feedback. More recently, agent-based systems have incorporated richer perception, reasoning, and interaction capabilities. Advances in representation learning and large language models enable agents to perform multi-step reasoning, interact with external tools, and execute complex workflows (Wei et al. 2022). Tool-augmented and reasoning-oriented agents have demonstrated strong performance in question answering, code generation, and planning by iteratively gathering information, invoking tools, and refining intermediate results (Yao et al. 2022). Meanwhile, learning-based decision-making has been explored in data-driven settings, including adaptive model selection, automated machine learning, and policy learning over structured action spaces (Baratchi et al. 2024). These methods highlight the potential of coordinating multiple components within a unified policy framework, but are mainly studied in domains with well-defined states, actions, and evaluation signals.

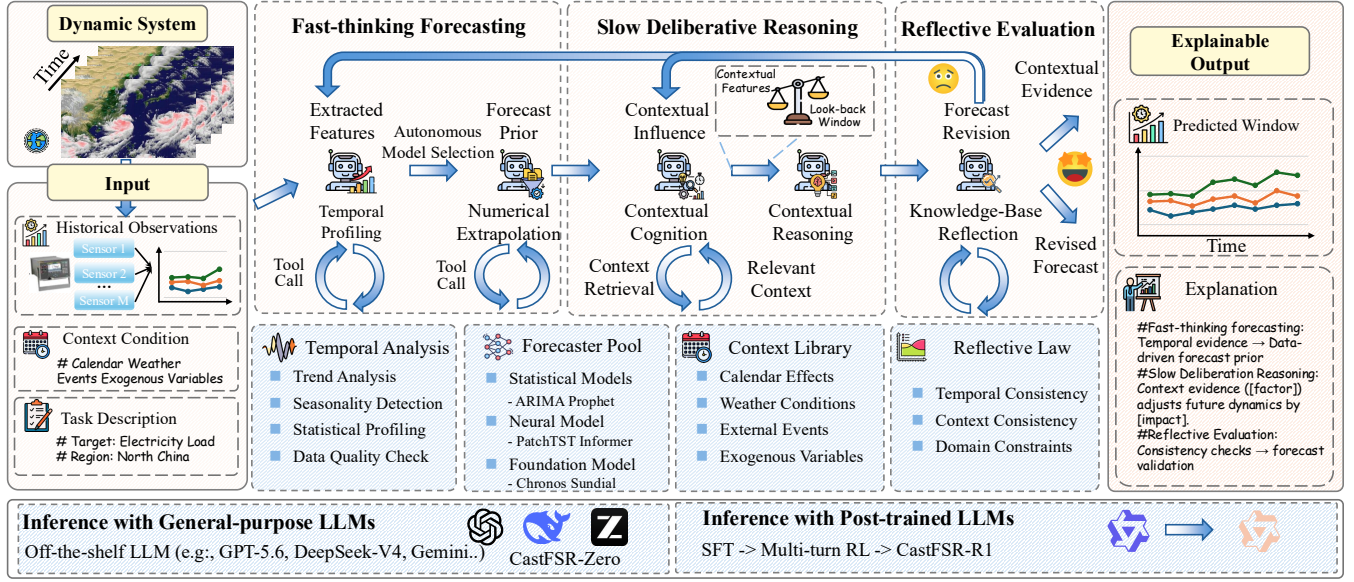


Figure 2: Overview of the CastFSR framework for agentic time series forecasting.

The Proposed CastFSR

Overview

As shown in Figure 2, CastFSR formulates context-aware TSF as a Fast–Slow–Reflect sequential decision-making process. A time series is viewed as an interface to dynamic systems, where future evolution is shaped by both historical patterns and contextual conditions. Fast-thinking forecasting constructs data-driven forecast priors using lightweight forecasters that capture temporal regularities, while slow deliberative reasoning leverages LLMs to retrieve contextual evidence and reason about its influence on future dynamics. Reflective evaluation further validates forecasts against temporal regularities and domain constraints. The same workflow is model-agnostic: it can be executed directly by off-the-shelf LLMs in a training-free setting or internalized by compact policy models through two-stage training.

Sequential Forecasting Formulation

We formulate TSF as a *sequential decision-making problem*, where forecasting requires a sequence of decisions, including understanding temporal dynamics, selecting numerical priors, retrieving contextual evidence, reasoning about future impacts, and validating predictions. Let $\mathcal{I} = (\mathcal{D}_{\text{task}}, \mathcal{D}_{\text{domain}}, \mathbf{X}_{1:L}, \mathcal{H}_{\text{ctx}})$ denote the input, where $\mathcal{D}_{\text{task}}$ defines the forecasting target and horizon, $\mathcal{D}_{\text{domain}}$ provides domain attributes, $\mathbf{X}_{1:L}$ represents the historical look-back window, and $\mathcal{H}_{\text{ctx}}$ denotes available contextual histories or future-known contextual resources. The objective is to predict future values $\mathbf{Y}_{1:H}$ and generate $\hat{\mathbf{Y}}_{1:H}$. Instead of treating forecasting as a direct mapping from history to future values, we represent the forecasting workflow as a policy π_{θ} that interacts with a forecasting environment for at most K steps. At step k , the agent observes a state:

$$s_k = \Phi(\mathcal{I}, \mathcal{M}_k, \hat{\mathbf{Y}}_k^{\text{prior}}, \mathcal{C}_k), \quad (1)$$

where $\mathcal{M}_k$ denotes the maintained memory state, $\hat{\mathbf{Y}}_k^{\text{prior}}$ is the current forecast prior, and $\mathcal{C}_k$ represents retrieved contextual evidence. The agent then selects an action:

$$a_k \sim \pi_{\theta}(\cdot | s_k), \quad a_k \in \mathcal{A}, \quad (2)$$

where $\mathcal{A} = \mathcal{A}_{\text{fast}} \cup \mathcal{A}_{\text{slow}} \cup \mathcal{A}_{\text{reflect}} \cup \mathcal{A}_{\text{final}}$ includes fast-thinking forecasting, slow deliberative reasoning, reflective evaluation, and final-answer actions.

Fast-thinking Forecasting

The fast-thinking stage constructs a pattern-based forecast prior from the look-back window. It operationalizes numerical extrapolation: if past observations encode stable regularities of the underlying system, then specialized forecasting models can quickly exploit these inductive biases without expensive semantic reasoning. This stage combines temporal pattern analysis with adaptive numerical extrapolation, allowing CastFSR to use mature forecasters for value generation while keeping the LLM responsible for orchestration rather than unconstrained numerical generation. CastFSR is equipped with a modular toolkit:

$$\mathcal{T} = \mathcal{T}_{\text{feat}} \cup \mathcal{T}_{\text{pred}}, \quad (3)$$

where $\mathcal{T}_{\text{feat}}$ contains feature-extraction tools and $\mathcal{T}_{\text{pred}}$ contains forecasting models. In the feature-extraction phase, the agent invokes complementary tools to characterize the input series. Specifically, trend analysis extracts temporal evolution patterns, seasonality detection captures periodic behaviors, statistical profiling summarizes distributional properties, and data-quality analysis identifies irregularities. These features provide structured evidence for model selection and prevent decisions based solely on raw observations. After feature extraction, CastFSR selects a forecasting model $m \in \mathcal{P}$ through a unified prediction interface, where $\mathcal{P}$ includes statistical

models, deep models, and foundation models. The selected predictor generates a forecast prior:

$$\hat{\mathbf{Y}}^{\text{prior}} = T_{\text{pred}}(\mathbf{X}_{1:L}; m), \quad m \sim \pi_{\theta}(\cdot | s_k). \quad (4)$$

By treating predictors as tools rather than fixed components, CastFSR adaptively routes different inputs to models and constructs data-driven priors from historical patterns.

Slow Deliberative Reasoning

Slow deliberative reasoning examines whether the pattern-based prior remains reliable under future-relevant contextual conditions. It models context-aware temporal reasoning by recognizing that future dynamics may deviate from historical extrapolation due to external factors such as weather, calendar effects, and operational constraints. Unlike fast-thinking forecasting, slow deliberative reasoning selectively reasons about when and how contexts influence future evolution rather than replacing all forecasts. It identifies relevant factors, determines retrieval ranges, and estimates their impacts on the forecast prior. Given the current memory, temporal features, domain description, and forecast prior, the agent performs contextual cognition to retrieve horizon-aligned evidence. Instead of using a fixed look-back window, CastFSR adaptively searches long-range contextual histories and selects context-specific windows, as different factors operate at different temporal scales. The retrieved evidence is then used to reason about future changes and refine the prior:

$$\mathcal{C}_k = T_{\text{ctx}}(\mathcal{D}_{\text{task}}, \mathcal{D}_{\text{domain}}, \mathbf{X}_{1:L}, \mathcal{H}_{\text{ctx}}, \hat{\mathbf{Y}}^{\text{prior}}, \mathcal{M}_k; \mathcal{W}_k), \quad (5)$$

where T_{ctx} denotes context retrieval and $\mathcal{W}_k$ contains adaptive look-back windows for different context types. The output $\mathcal{C}_k$ is stored as structured evidence.

After contextual cognition, CastFSR estimates the contextual influence by determining whether retrieved evidence is relevant to the forecast horizon, how it changes future dynamics, and which parts of the numerical prior should be revised. If the evidence is weak, redundant with historical patterns, or outside the effective context range, the agent preserves the original prior. Otherwise, it reasons about the direction, magnitude, and temporal scope of the contextual influence. The agent then performs *contextual reasoning* to transform impact assessment into a candidate forecast. Instead of generating values from scratch, the deliberation module refines the numerical prior by applying context-grounded adjustments to relevant timestamps or segments:

$$\hat{\mathbf{Y}}^{\text{cand}} = G_{\theta}(\hat{\mathbf{Y}}^{\text{prior}}, \mathcal{C}_k, \mathcal{M}_k), \quad (6)$$

where G_{θ} denotes the deliberation process. This design couples intrinsic temporal dynamics with contextual information: the prior provides numerical stability, while retrieved evidence determines when and where deviations from extrapolation are needed. The resulting candidate forecast, together with contextual evidence and revision rationale, is passed to reflective evaluation.

Reflective Evaluation

The reflective evaluation stage verifies the candidate forecast before final output. It enforces the constraint-aware principle

that forecasts should be not only accurate but also consistent with temporal regularities, domain knowledge, and operational constraints. The agent evaluates $\hat{\mathbf{Y}}^{\text{cand}}$ through three consistency checks. Temporal consistency examines trends, seasonal patterns, turning points, and continuity with historical observations. Contextual consistency verifies whether forecast revisions are supported by retrieved evidence and whether their timing and direction align with selected context windows. Domain consistency checks validity constraints, including units, non-negativity, capacity limits, timestamp correctness, and task-specific requirements. The evaluation process produces:

$$\mathcal{E}_k = E_{\theta}(\hat{\mathbf{Y}}^{\text{cand}}, \hat{\mathbf{Y}}^{\text{prior}}, \mathcal{C}_k, \mathcal{M}_k). \quad (7)$$

Based on $\mathcal{E}_k$, the agent decides whether to accept or refine the candidate forecast. When inconsistencies are localized, reflective evaluation applies targeted corrections rather than regenerating the entire horizon, preserving reliable predictions while repairing invalid segments. A forecast is accepted only after satisfying temporal and format requirements:

$$\hat{\mathbf{Y}}_{1:H} = F_{\theta}(\hat{\mathbf{Y}}^{\text{cand}}, \mathcal{E}_k, \mathcal{M}_k). \quad (8)$$

The final output includes both the predicted window and an interpretable trajectory report describing the evidence used, constraints checked, and revision rationale.

Model-Agnostic Instantiation of CastFSR

The workflow separates forecasting logic from the policy that coordinates it. In the training-free setting, an off-the-shelf LLM follows the stage-constrained action interface and directly orchestrates feature profiling, model selection, context retrieval, and reflective evaluation. This mode requires no task-specific parameter update, has no dependency on a particular backbone model, and permits different general-purpose LLMs to instantiate the same framework.

For privacy-sensitive and cost-constrained deployment, relying on large proprietary models may be undesirable. We therefore further explore whether compact models can acquire the same workflow-level capability. Specifically, we internalize the workflow in a small model through two-stage training. First, SFT learns executable Fast–Slow–Reflect behavior from general LLM trajectories distilled on the full cross-domain training data. The validation removes trajectories with illegal tool calls, incorrect stage order, ground-truth leakage, invalid timestamps, or incomplete forecasts. Second, multi-turn RL optimizes complete decision trajectories under delayed forecasting feedback. For a group of trajectories $\{\tau_i\}_{i=1}^G$, we normalize their episode rewards as:

$$A_i = \frac{R_i - \mu_R}{\sigma_R + \epsilon}, \quad (9)$$

and optimize the clipped GRPO objective (Shao et al. 2024b). The episode reward combines output validity and numerical accuracy with structural agreement on trend, seasonality, and change points across forecast horizons. SFT therefore establishes reliable orchestration, while RL improves context selection, revision, and reflective evaluation decisions toward end-to-end forecasting quality.

Model	Long-term forecasting										Short-term forecasting									
	ETTh1		ETTh2		ETTm1		ETTm2		Wind		BE		DE		FR		NP		PJM	
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ARIMA	0.110	0.243	0.443	0.505	0.055	<u>0.169</u>	0.376	0.429	3.682	1.361	1.978	0.814	3.269	1.076	6.410	1.189	1.798	0.801	0.907	0.710
Prophet	0.630	0.546	0.942	0.710	0.091	0.216	0.216	0.313	7.874	2.094	2.321	0.918	2.103	0.986	6.831	1.280	0.707	0.583	0.410	0.500
PatchTST	0.114	0.247	0.251	0.375	0.061	0.184	0.133	0.245	2.556	1.212	1.905	0.647	0.983	0.666	5.021	0.740	0.454	0.441	0.190	0.323
iTransformer	0.102	0.243	0.257	0.390	0.058	0.180	0.138	0.256	2.657	1.196	1.684	0.569	1.298	0.697	6.066	0.869	0.373	0.404	0.238	0.344
ConvTimeNet	0.097	0.236	0.252	0.380	<u>0.056</u>	0.174	0.134	0.247	2.544	1.206	1.482	0.581	0.964	0.665	5.376	0.800	0.393	0.411	0.226	0.342
TimeXer	0.108	0.248	0.255	0.386	0.058	0.185	0.136	0.249	2.479	1.214	1.437	0.502	1.009	0.708	5.578	0.827	0.412	0.416	0.197	0.318
DLinear	0.103	0.249	0.252	0.386	0.061	0.184	0.138	0.253	2.441	1.149	1.697	0.677	1.000	0.699	6.369	1.000	0.431	0.446	0.286	0.387
TimesFM	0.104	0.237	0.266	0.388	0.059	0.176	0.353	0.412	3.171	1.273	1.349	0.510	0.967	0.660	4.580	0.651	0.317	0.361	0.207	0.324
Sundial	0.106	0.247	0.273	0.391	0.058	0.172	0.213	0.331	2.780	1.244	1.478	0.513	1.155	0.700	6.105	0.682	0.362	0.378	0.201	0.324
OFA	0.100	0.242	<u>0.247</u>	0.378	0.058	0.179	0.127	0.242	2.708	1.250	1.522	0.610	1.065	0.699	5.659	0.868	0.383	0.420	0.252	0.369
Time-LLM	0.097	0.240	0.250	0.378	0.059	0.183	0.138	0.254	2.648	1.230	1.758	0.701	1.025	0.707	6.450	0.971	0.422	0.447	0.253	0.368
PromptCast	0.123	0.322	0.299	0.372	0.085	0.241	0.149	0.296	4.404	1.200	1.870	0.637	1.229	0.708	7.302	1.173	0.565	0.449	0.316	0.454
TokenCast	0.121	0.277	0.385	0.472	0.055	0.177	0.137	0.259	2.926	1.265	1.485	0.547	0.973	0.702	6.363	0.961	0.412	0.411	0.215	0.319
S ² IP-LLM	0.119	0.277	0.290	0.424	<u>0.056</u>	0.176	<u>0.129</u>	0.249	2.638	1.184	1.591	0.567	1.315	0.722	5.765	0.888	0.389	0.391	0.237	0.339
TimeReasoner	0.126	0.283	0.375	0.425	0.072	0.205	0.173	0.295	3.553	1.390	1.299	0.466	1.082	0.760	4.644	0.650	0.368	0.386	0.207	0.328
TimeSeriesScientist	0.213	0.290	0.498	0.505	0.092	0.205	0.150	0.300	5.907	1.680	2.057	0.712	1.028	0.673	5.977	1.033	0.606	0.476	0.302	0.439
AlphaCast	0.123	0.225	0.279	0.372	0.075	0.185	0.131	0.271	2.375	0.856	1.484	0.510	1.043	0.662	5.318	0.740	0.316	0.365	0.176	0.292
CastFSR-Zero	<u>0.081</u>	<u>0.212</u>	0.268	<u>0.361</u>	0.055	0.168	0.127	<u>0.238</u>	1.596	<u>0.886</u>	<u>0.960</u>	<u>0.386</u>	<u>0.393</u>	<u>0.408</u>	3.939	0.501	<u>0.221</u>	<u>0.277</u>	<u>0.140</u>	<u>0.268</u>
CastFSR-R1	0.077	0.210	0.242	0.351	0.055	0.168	0.127	0.237	<u>1.757</u>	0.918	0.927	0.375	0.386	0.405	<u>4.032</u>	<u>0.502</u>	0.172	0.247	0.133	0.260

Table 1: Overall forecasting performance on benchmark datasets. Lower values indicate better performance. The best results are highlighted in bold, and the second-best are underlined.

Setting	Dataset	Domain	Length	Variables	Frequency
Long-term	ETTh1 & ETTh2	Electricity	17,420	7	1 hour
	ETTm1 & ETTm2	Electricity	69,680	7	15 mins
	Wind	Energy	48,673	7	15 mins
Short-term	BE	Energy	14,496	3	1 hour
	DE	Energy	14,496	3	1 hour
	FR	Energy	14,496	3	1 hour
	NP	Energy	14,496	3	1 hour
	PJM	Energy	14,496	3	1 hour

Table 2: Statistics of diverse real-world time series datasets.

Experiments

In this section, we first introduce the experimental settings and then present experimental results, ablation studies, and case analyses to evaluate the proposed framework.

Experimental Settings

Datasets. Table 2 summarizes the real-world datasets, covering diverse domains, resolutions, and horizons. The ETT benchmark (Zhou et al. 2021) contains electricity transformer measurements at hourly and 15-minute resolutions with long-range dependencies. The Wind dataset follows the benchmark used in prior generative forecasting studies (Li et al. 2022). For short-term forecasting, datasets are hourly electricity price datasets from the EPF benchmark (Lago et al. 2021), representing regional power markets.

Baselines. We compare CastFSR against baselines in Table 1. Statistical methods include ARIMA (Hyndman and Khandakar 2008) and Prophet (Taylor and Letham 2018), which capture temporal patterns through classical modeling. Deep learning approaches include DLinear (Zeng et al. 2023), ConvTimeNet (Cheng et al. 2025b), PatchTST (Nie et al. 2023), iTransformer (Liu et al. 2024), and TimeXer (Wang et al. 2024), covering MLP-, CNN-, and Transformer-based architectures. Foundation models include TimesFM (Das et al. 2024) and Sundial (Liu et al. 2025b).

LLM-based methods include OFA (Zhou et al. 2023), Time-LLM (Jin et al. 2024), TokenCast (Tao et al. 2025), S²IP-LLM (Pan et al. 2024), TimeReasoner (Cheng et al. 2026), and PromptCast (Xue and Salim 2023), adapting LLMs via alignment, tokenization, prompting, and reasoning. We further compare agentic systems TimeSeriesScientist (Zhao et al. 2025) and AlphaCast (Zhang et al. 2025).

Implementation Details. For training-free inference, we instantiate CastFSR-Zero, where DeepSeek V4 Flash (Xu et al. 2026) serves as the reasoning engine. We further develop CastFSR-R1 by fine-tuning Qwen3-4B (Bai et al. 2023) with SFT and RL on distilled cross-domain trajectories, using 16 Ascend NPUs. Deep learning baselines follow their official configurations. All methods adopt unified forecasting settings: long-term tasks use a look-back/horizon of 96, while short-term tasks use 168/24. We report MSE and MAE as evaluation metrics. Detailed dataset, baseline, and implementation descriptions are in the Appendix.

Main Results

Table 1 compares CastFSR with forecasting baselines across long- and short-term benchmarks. CastFSR achieves the best or second-best performance on most metrics, demonstrating the effectiveness of the Fast-Slow-Reflect workflow for context-aware forecasting. CastFSR-R1 further improves over CastFSR-Zero on most benchmarks, indicating that SFT and RL help compact models internalize the Fast-Slow-Reflect reasoning process. Meanwhile, CastFSR-Zero remains competitive, showing that the workflow performs well with off-the-shelf LLMs without task-specific optimization. Compared with existing LLM-based and agentic methods, CastFSR benefits from adaptive context utilization beyond direct prompting. These results demonstrate that combining fast-thinking forecasting, slow deliberative reasoning, and reflective evaluation enables accurate context-aware forecasting across diverse temporal domains.

Variant	ETTh1		ETTh2		ETTm1		ETTm2		Wind		BE		DE		FR		NP		PJM	
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
w/o Fast-thinking	0.089	0.224	0.339	0.403	0.061	0.184	0.176	0.283	2.403	1.111	1.225	0.455	0.481	0.451	4.251	0.534	0.334	0.336	0.168	0.300
w/o Slow Deliberative Reasoning	<u>0.086</u>	<u>0.216</u>	<u>0.298</u>	<u>0.373</u>	0.055	<u>0.169</u>	0.157	0.265	2.310	1.054	1.147	0.431	0.438	0.426	3.936	<u>0.504</u>	0.292	<u>0.301</u>	<u>0.149</u>	<u>0.273</u>
w/o Reflective Evaluation	<u>0.086</u>	<u>0.216</u>	0.320	0.388	0.055	<u>0.169</u>	<u>0.139</u>	<u>0.255</u>	<u>2.032</u>	<u>0.997</u>	<u>1.011</u>	<u>0.398</u>	<u>0.421</u>	<u>0.416</u>	3.952	0.505	<u>0.284</u>	0.306	0.150	0.274
CastFSR-Zero	0.081	0.212	0.268	0.361	0.055	0.168	0.127	0.238	1.596	0.886	0.960	0.386	0.393	0.408	<u>3.939</u>	0.501	0.221	0.277	0.140	0.268

Table 3: Ablation study of Fast–Slow–Reflect modules across benchmark datasets. We compare CastFSR-Zero against variants removing Fast-thinking Forecasting, Slow Deliberative Reasoning, or Reflective Evaluation.

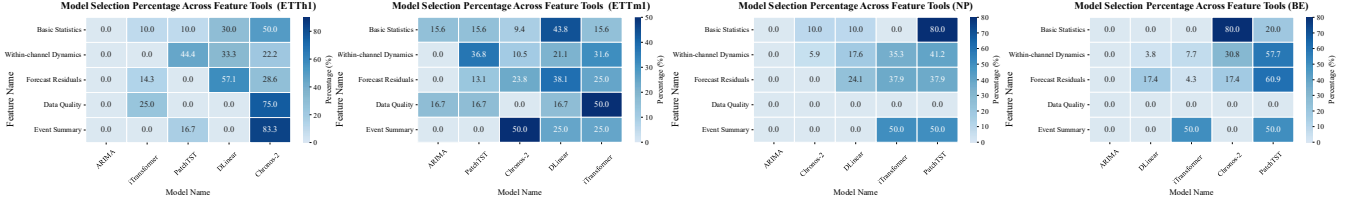


Figure 3: Feature-conditioned model selection patterns. Model-selection frequencies (%) conditioned on diagnostic features. Rows denote feature tools and columns denote candidate models; darker cells indicate routing frequency.

Strategy	ETTh1		ETTm1		BE		NP	
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Best Model Result	0.916	0.213	0.063	0.305	1.164	0.496	0.386	0.046
CastFSR Selection	0.890	0.225	0.057	0.176	1.099	0.448	0.338	0.043
Best Model	iTransformer		TimeXer		PatchTST		PatchTST	

Table 4: Comparison between the best fixed model in the candidate pool and feature-conditioned model selection.

Ablation Study of the Workflow Modules

Table 3 evaluates each component by removing fast-thinking forecasting, slow deliberative reasoning, or reflective evaluation from CastFSR-Zero. Removing a component generally degrades performance across metrics, confirming their complementary roles in context-aware forecasting. Removing fast-thinking forecasting causes the largest degradation, showing that data-driven forecast priors provide the foundation for subsequent reasoning. Without this prior, the agent relies more on LLM generation and produces less accurate forecasts. Removing slow deliberative reasoning also reduces performance on most metrics, highlighting the importance of contextual evidence retrieval and reasoning. Removing reflective evaluation further validates the necessity of temporal consistency and domain constraint checks.

Exploration Analysis of Model Selection

Figure 3 illustrates the feature-conditioned model selection patterns of CastFSR. The routing distributions show that CastFSR dynamically selects forecasting experts based on temporal characteristics rather than a fixed predictor. Different features induce distinct preferences, indicating learned associations between series properties and forecasting strategies. Trend- and residual-related features favor models with stronger temporal modeling capabilities, while others match underlying dynamics. Adaptive routing constructs forecast priors by leveraging complementary experts, validating fast-thinking forecasting. Furthermore, Table 4 compares selected

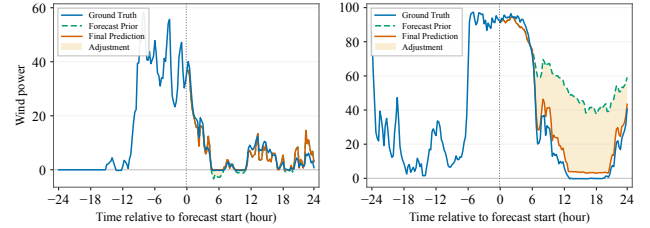


Figure 4: Analysis of adaptive slow deliberative reasoning, where CastFSR effectively balances the influence of historical patterns and contextual features.

forecasters with the best fixed model. CastFSR selection improves most metrics over fixed experts, demonstrating the benefit of adaptive model selection under diverse dynamics.

Analyzing Adaptive Slow Deliberative Reasoning

Figure 4 illustrates how CastFSR adaptively refines forecast priors by balancing historical patterns and contextual features. The forecast prior provides a data-driven prior from historical observations, while CastFSR evaluates contextual evidence and selectively determines its influence. When historical patterns remain reliable, CastFSR preserves the prior with limited adjustments. When contextual signals reveal meaningful deviations, it highlights relevant contextual effects and performs targeted revisions. These adjustments demonstrate that CastFSR does not blindly follow external contexts, but adaptively weighs historical consistency and contextual relevance to refine forecasts. The results validate the effectiveness of contextual cognition in guiding adaptive forecast refinement under dynamic forecasting scenarios.

Impact of Pre-training LLMs

Table 5 evaluates CastFSR-Zero with different pretrained LLMs. CastFSR-Zero maintains competitive performance

LLM Coordinator	ETTh1		ETTm1		DE		NP	
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
GPT-5.6-sol	<u>0.081</u>	0.213	<u>0.055</u>	0.168	0.400	0.414	0.218	<u>0.266</u>
LongCat-2.0	0.089	0.219	0.056	0.168	0.383	<u>0.412</u>	0.221	0.271
DeepSeek V4 Pro	<u>0.081</u>	<u>0.212</u>	0.054	0.165	0.403	0.417	<u>0.216</u>	0.267
DeepSeek V4 Flash	<u>0.081</u>	<u>0.212</u>	<u>0.055</u>	0.168	<u>0.393</u>	0.408	0.221	0.277
GLM-5.2	0.080	0.210	<u>0.055</u>	<u>0.167</u>	0.405	0.427	0.207	0.260

Table 5: Training-free forecasting performance of CastFSR-Zero with different Pre-training LLMs.

Variant	ETTh1		ETTm1		DE		NP	
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
w/o SFT	<u>0.081</u>	<u>0.213</u>	0.055	0.168	0.435	0.436	0.237	0.280
w/o RL	0.082	0.215	0.055	<u>0.170</u>	<u>0.392</u>	<u>0.409</u>	<u>0.217</u>	<u>0.268</u>
CastFSR-R1	0.077	0.210	0.055	0.168	0.386	0.405	0.172	0.247

Table 6: Performance comparison of CastFSR-R1 with variants removing supervised fine-tuning (w/o SFT) or reinforcement learning (w/o RL).

across various backbones, demonstrating the generalizability of the proposed workflow. No single LLM consistently dominates, indicating that forecasting performance depends on not only language capability but also tool coordination and evidence interpretation. Although GPT-5.6-sol exhibits strong reasoning ability, its performance varies across forecasting scenarios, where more conservative adjustments may limit contextual evidence utilization. We adopt DeepSeek V4 Flash as the default coordinator due to its favorable performance–cost trade-off, enabling efficient inference while maintaining competitive reasoning capability.

Effect of Training Strategies

Table 6 evaluates the contribution of SFT and RL by removing each training stage from CastFSR-R1. Removing either stage generally degrades forecasting performance on the reported metrics, with occasional ties on individual metrics, demonstrating that SFT and RL provide complementary benefits. Without SFT, the model lacks sufficient initialization for following the Fast–Slow–Reflect workflow. Removing RL also causes noticeable performance drops, indicating that reinforcement learning further improves adaptive decision-making and optimizes the coordination among forecasting priors, contextual evidence, and reflective evaluation. Overall, the full CastFSR-R1 achieves the best or tied-best performance, validating the effectiveness of the two-stage learning strategy for internalizing agentic forecasting capabilities.

Analysis of Context Utilization

Figure 5 compares CastFSR with a variant without contextual cognition. The performance degradation after removing context utilization demonstrates the importance of slow deliberative reasoning in identifying and incorporating external evidence. Without contextual reasoning, the model relies primarily on historical patterns and produces less accurate forecasts, especially under complex temporal dynamics. This

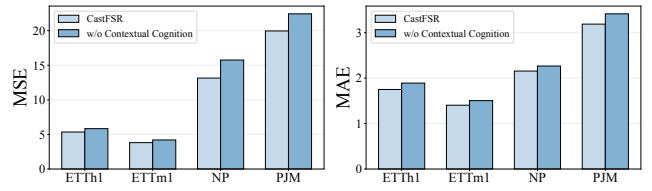


Figure 5: Analysis of context utilization strategies.

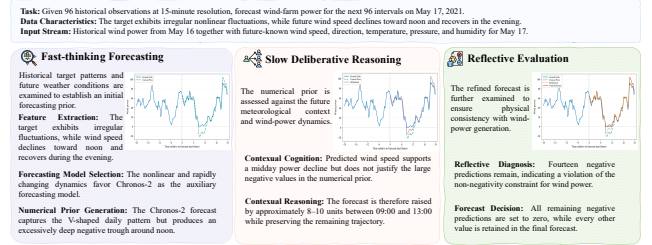


Figure 6: Case study of CastFSR for context-aware TSF.

confirms that effective context-aware forecasting requires not only numerical prediction but also adaptive interpretation of contextual signals. This benefit is evident when contextual shifts reshape future trajectories, particularly when abrupt events disrupt otherwise stable historical temporal patterns.

Case Study Analysis

Figure 6 presents a wind power case of CastFSR. During fast-thinking forecasting, CastFSR extracts temporal patterns and constructs a data-driven forecast prior. The selected expert captures the general trajectory but shows deviations under future wind-speed changes, motivating contextual reasoning. During slow deliberative reasoning, CastFSR evaluates contextual evidence and identifies that wind conditions support higher power generation. Instead of replacing the prior, it adjusts the forecast based on contextual relevance. Finally, reflective evaluation checks domain constraints and corrects implausible negative predictions. This case demonstrates that CastFSR integrates numerical priors, contextual reasoning, and domain-aware evaluation to produce accurate and reliable time series forecasting.

Conclusion

In this work, we proposed CastFSR, an agentic TSF framework that reformulates context-aware forecasting as a Fast–Slow–Reflect sequential decision-making process. By coordinating Fast-thinking Forecasting, Slow Deliberative Reasoning, and Reflective Evaluation, CastFSR offers a principled alternative to single-pass numerical extrapolation. The framework supports both training-free execution with off-the-shelf LLMs and compact deployment through supervised fine-tuning and multi-turn reinforcement learning. Extensive experiments on real-world datasets demonstrate that the proposed workflow improves forecasting accuracy across diverse domains, validating the complementary roles of data-driven prior construction, context-aware temporal reasoning, and constraint-aware forecast validation.

References

- Ansari, A. F.; Shchur, O.; Küken, J.; Auer, A.; Han, B.; Mercado, P.; Rangapuram, S. S.; Shen, H.; Stella, L.; Zhang, X.; et al. 2025. Chronos-2: From univariate to universal forecasting. *arXiv preprint arXiv:2510.15821*.
- Ashok, A.; Williams, A. R.; Marcotte, É.; Zantedeschi, V.; Subramanian, J.; Riachi, R.; Requeima, J.; Lacoste, A.; Rish, I.; Chapados, N.; et al. 2024. Context is key: a benchmark for forecasting with essential textual information. In *NeurIPS Workshop on Time Series in the Age of Large Models*.
- Bai, J.; Bai, S.; Chu, Y.; Cui, Z.; Dang, K.; Deng, X.; Fan, Y.; Ge, W.; Han, Y.; Huang, F.; et al. 2023. Qwen technical report. *arXiv preprint arXiv:2309.16609*.
- Baratchi, M.; Wang, C.; Limmer, S.; Van Rijn, J. N.; Hoos, H.; Bäck, T.; and Olhofer, M. 2024. Automated machine learning: past, present and future. *Artificial intelligence review*, 57(5): 122.
- Box, G. E. P.; Jenkins, G. M.; Reinsel, G. C.; and Ljung, G. M. 2015. *Time Series Analysis: Forecasting and Control*. Wiley, 5 edition.
- Chang, C.; Shi, Y.; Cao, D.; Yang, W.; Hwang, J.; Wang, H.; Pang, J.; Wang, W.; Liu, Y.; Peng, W.-C.; et al. 2025. A survey of reasoning and agentic systems in time series with large language models. *arXiv preprint arXiv:2509.11575*.
- Cheng, M.; Liu, Z.; Tao, X.; Liu, Q.; Zhang, J.; Pan, T.; Zhang, S.; He, P.; Zhang, X.; Wang, D.; et al. 2025a. A Comprehensive Survey of Time Series Forecasting: Concepts, Challenges, and Future Directions. *Authorea Preprints*.
- Cheng, M.; Wang, J.; Wang, D.; Tao, X.; Liu, Q.; and Chen, E. 2026. Can slow-thinking llms reason over time? empirical studies in time series forecasting. In *Proceedings of the Nineteenth ACM International Conference on Web Search and Data Mining*, 99–110.
- Cheng, M.; Yang, J.; Pan, T.; Liu, Q.; Li, Z.; and Wang, S. 2025b. Convtimenet: A deep hierarchical fully convolutional model for multivariate time series analysis. In *Companion Proceedings of the ACM on Web Conference 2025*, 171–180.
- Das, A.; Kong, W.; Sen, R.; and Zhou, Y. 2024. A decoder-only foundation model for time-series forecasting. In *Forty-first International Conference on Machine Learning*.
- Deng, J.; Chen, X.; Jiang, R.; Song, X.; and Tsang, I. W. 2022. A multi-view multi-task learning framework for multivariate time series forecasting. *IEEE Transactions on Knowledge and Data Engineering*, 35(8): 7665–7680.
- Gao, C.; Lan, X.; Li, N.; Yuan, Y.; Ding, J.; Zhou, Z.; Xu, F.; and Li, Y. 2024. Large language models empowered agent-based modeling and simulation: A survey and perspectives. *Humanities and Social Sciences Communications*, 11(1): 1–24.
- Guan, T.; Meng, Z.; Li, D.; Wang, S.; Yang, C.-H. H.; Wen, Q.; Liu, Z.; Siniscalchi, S. M.; Jin, M.; and Pan, S. 2026. TimeOmni-1: Incentivizing Complex Reasoning with Time Series in Large Language Models. In *The Fourteenth International Conference on Learning Representations*.
- Guo, S.; Lin, Y.; Feng, N.; Song, C.; and Wan, H. 2019. Attention based spatial-temporal graph convolutional networks for traffic flow forecasting. In *Proceedings of the AAAI conference on artificial intelligence*, volume 33, 922–929.
- Holland, J. H. 1992. Complex adaptive systems. *Daedalus*, 121(1): 17–30.
- Hyndman, R. J.; and Khandakar, Y. 2008. Automatic time series forecasting: the forecast package for R. *Journal of statistical software*, 27: 1–22.
- Ji, J.; Wang, J.; Huang, C.; Wu, J.; Xu, B.; Wu, Z.; Zhang, J.; and Zheng, Y. 2023. Spatio-temporal self-supervised learning for traffic flow prediction. In *Proceedings of the AAAI conference on artificial intelligence*, volume 37, 4356–4364.
- Jia, F.; Wang, K.; Zheng, Y.; Cao, D.; and Liu, Y. 2024. Gpt4mts: Prompt-based large language model for multi-modal time-series forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, 23343–23351.
- Jin, G.; Liang, Y.; Fang, Y.; Shao, Z.; Huang, J.; Zhang, J.; and Zheng, Y. 2023. Spatio-temporal graph neural networks for predictive learning in urban computing: A survey. *IEEE transactions on knowledge and data engineering*, 36(10): 5388–5408.
- Jin, M.; Wang, S.; Ma, L.; Chu, Z.; Zhang, J. Y.; Shi, X.; Chen, P.-Y.; Liang, Y.; Li, Y.-F.; Pan, S.; et al. 2024. Time-LLM: Time Series Forecasting by Reprogramming Large Language Models. In *The Twelfth International Conference on Learning Representations*.
- Lago, J.; Marcjasz, G.; De Schutter, B.; and Weron, R. 2021. Forecasting day-ahead electricity prices: A review of state-of-the-art algorithms, best practices and an open-access benchmark. *Applied Energy*, 293: 116983.
- Li, Y.; Lu, X.; Wang, Y.; and Dou, D. 2022. Generative time series forecasting with diffusion, denoise, and disentanglement. *Advances in Neural Information Processing Systems*, 35: 23009–23022.
- Liu, C.; Xu, Q.; Miao, H.; Yang, S.; Zhang, L.; Long, C.; Li, Z.; and Zhao, R. 2025a. Timecma: Towards llm-empowered multivariate time series forecasting via cross-modality alignment. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, 18780–18788.
- Liu, Y.; Hu, T.; Zhang, H.; Wu, H.; Wang, S.; Ma, L.; and Long, M. 2024. iTransformer: Inverted Transformers Are Effective for Time Series Forecasting. In *The Twelfth International Conference on Learning Representations*.
- Liu, Y.; Qin, G.; Shi, Z.; Chen, Z.; Yang, C.; Huang, X.; Wang, J.; and Long, M. 2025b. Sundial: A Family of Highly Capable Time Series Foundation Models. In *International Conference on Machine Learning*, 39295–39317. PMLR.
- Lu, J.; Chen, P.; Guo, C.; Shu, Y.; Wang, M.; and Yang, B. 2026. Towards non-stationary time series forecasting with temporal stabilization and frequency differencing. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, 24070–24078.
- Luo, Y.; Zhou, Y.; Cheng, M.; Wang, J.; Wang, D.; Pan, T.; and Zhang, J. 2025. Time Series Forecasting as Reasoning: A Slow-Thinking Approach with Reinforced LLMs. *arXiv preprint arXiv:2506.10630*.

- Masini, R. P.; Medeiros, M. C.; and Mendes, E. F. 2023. Machine learning advances for time series forecasting. *Journal of economic surveys*, 37(1): 76–111.
- Nie, Y.; H. Nguyen, N.; Sinthong, P.; and Kalagnanam, J. 2023. A Time Series is Worth 64 Words: Long-term Forecasting with Transformers. In *International Conference on Learning Representations*.
- Palaskar, S.; Ekambaram, V.; Jati, A.; Gantayat, N.; Saha, A.; Nagar, S.; Nguyen, N. H.; Dayama, P.; Sindhgatta, R.; Mohapatra, P.; et al. 2024. Automixer for improved multivariate time-series forecasting on business and it observability data. In *Proceedings of the AAAI conference on artificial intelligence*, volume 38, 22962–22968.
- Pan, Z.; Jiang, Y.; Garg, S.; Schneider, A.; Nevmyvaka, Y.; and Song, D. 2024. S²IP-LLM: Semantic Space Informed Prompt Learning with LLM for Time Series Forecasting. In *Forty-first International Conference on Machine Learning*.
- Shao, Z.; Wang, F.; Xu, Y.; Wei, W.; Yu, C.; Zhang, Z.; Yao, D.; Sun, T.; Jin, G.; Cao, X.; et al. 2024a. Exploring progress in multivariate time series forecasting: Comprehensive benchmarking and heterogeneity analysis. *IEEE Transactions on Knowledge and Data Engineering*, 37(1): 291–305.
- Shao, Z.; Wang, P.; Zhu, Q.; Xu, R.; Song, J.; Bi, X.; Zhang, H.; Zhang, M.; Li, Y.; Wu, Y.; et al. 2024b. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*.
- Shi, X.; Wang, S.; Nie, Y.; Li, D.; Ye, Z.; Wen, Q.; and Jin, M. 2025. Time-MoE: Billion-Scale Time Series Foundation Models with Mixture of Experts. In *The Thirteenth International Conference on Learning Representations*.
- Tao, X.; Cheng, M.; Guo, Z.; Yu, S.; Liu, Y.; Liu, Q.; and Wang, S. 2026. Memcast: Memory-driven time series forecasting with experience-conditioned reasoning. *arXiv preprint arXiv:2602.03164*.
- Tao, X.; Zhang, S.; Cheng, M.; Wang, D.; Pan, T.; Pan, B.; Zhang, C.; and Wang, S. 2025. From values to tokens: An llm-driven framework for context-aware time series forecasting via symbolic discretization. *arXiv preprint arXiv:2508.09191*.
- Taylor, S. J.; and Letham, B. 2018. Forecasting at scale. *The American Statistician*, 72(1): 37–45.
- Wang, F.; Zhang, Z.; Zhang, X.; Wu, Z.; Mo, T.; Lu, Q.; Wang, W.; Li, R.; Xu, J.; Tang, X.; et al. 2025. A comprehensive survey of small language models in the era of large language models: Techniques, enhancements, applications, collaboration with llms, and trustworthiness. *ACM Transactions on Intelligent Systems and Technology*, 16(6): 1–87.
- Wang, Y.; Smola, A.; Maddix, D.; Gasthaus, J.; Foster, D.; and Januschowski, T. 2019. Deep factors for forecasting. In *International conference on machine learning*, 6607–6617. PMLR.
- Wang, Y.; Wu, H.; Dong, J.; Qin, G.; Zhang, H.; Liu, Y.; Qiu, Y.; Wang, J.; and Long, M. 2024. Timexer: Empowering transformers for time series forecasting with exogenous variables. *Advances in Neural Information Processing Systems*, 37: 469–498.
- Wei, J.; Wang, X.; Schuurmans, D.; Bosma, M.; Xia, F.; Chi, E.; Le, Q. V.; Zhou, D.; et al. 2022. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35: 24824–24837.
- Winters, P. R. 1960. Forecasting sales by exponentially weighted moving averages. *Management science*, 6(3): 324–342.
- Woo, G.; Liu, C.; Kumar, A.; Xiong, C.; Savarese, S.; and Sahoo, D. 2024. Unified training of universal time series forecasting transformers. In *Forty-first International Conference on Machine Learning*.
- Wu, H.; Xu, J.; Wang, J.; and Long, M. 2021. Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting. *Advances in neural information processing systems*, 34: 22419–22430.
- Wu, X.; Lu, J.; Li, Z.; Qiu, X.; Hu, J.; Guo, C.; Jensen, C. S.; and Yang, B. 2026. TimeART: Towards Agentic Time Series Reasoning via Tool-Augmentation. *arXiv preprint arXiv:2601.13653*.
- Xu, A.; Lin, B.; Xue, B.; Wang, B.; Xu, B.; Wu, B.; Zhang, B.; Lin, C.; Dong, C.; Ling, C.; et al. 2026. Deepseek-v4: Towards highly efficient million-token context intelligence. *arXiv preprint arXiv:2606.19348*.
- Xue, H.; and Salim, F. D. 2023. Promptcast: A new prompt-based learning paradigm for time series forecasting. *IEEE Transactions on Knowledge and Data Engineering*, 36(11): 6851–6864.
- Yao, S.; Zhao, J.; Yu, D.; Du, N.; Shafran, I.; Narasimhan, K. R.; and Cao, Y. 2022. React: Synergizing reasoning and acting in language models. In *The eleventh international conference on learning representations*.
- Zeng, A.; Chen, M.; Zhang, L.; and Xu, Q. 2023. Are transformers effective for time series forecasting? In *Proceedings of the AAAI conference on artificial intelligence*, volume 37, 11121–11128.
- Zhang, W.; Liu, H.; Liu, Y.; Zhou, J.; and Xiong, H. 2020. Semi-supervised hierarchical recurrent graph neural network for city-wide parking availability prediction. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, 1186–1193.
- Zhang, X.; Gao, T.; Cheng, M.; Pan, B.; Guo, Z.; Liu, Y.; and Tao, X. 2025. AlphaCast: A Human Wisdom-LLM Intelligence Co-Reasoning Framework for Interactive Time Series Forecasting. *arXiv preprint arXiv:2511.08947*.
- Zhao, H.; Zhang, X.; Wei, J.; Xu, Y.; He, Y.; Sun, S.; and You, C. 2025. Timeseriesscientist: A general-purpose ai agent for time series analysis. *arXiv preprint arXiv:2510.01538*.
- Zhou, H.; Zhang, S.; Peng, J.; Zhang, S.; Li, J.; Xiong, H.; and Zhang, W. 2021. Informer: Beyond efficient transformer for long sequence time-series forecasting. In *Proceedings of the AAAI conference on artificial intelligence*, volume 35, 11106–11115.
- Zhou, T.; Niu, P.; Sun, L.; Jin, R.; et al. 2023. One fits all: Power general time series analysis by pretrained lm. *Advances in neural information processing systems*, 36: 43322–43355.

Detailed Dataset Descriptions

We evaluate CastFSR on a diverse collection of real-world time series datasets, including long-term time series forecasting (LTSF) benchmarks and short-term electricity price forecasting (EPF) benchmarks.

Long-term Time Series Forecasting

For long-horizon forecasting tasks, we consider benchmarks with complex temporal dependencies and diverse temporal patterns under challenging forecasting conditions.

- **ETT (Electricity Transformer Temperature):** The ETT benchmark (Zhou et al. 2021) provides two years of electricity transformer measurements, with Oil Temperature (OT) as the forecasting target. As a key indicator of transformer operating conditions, OT exhibits complex temporal dependencies and long-term variations.
 - **ETTh1:** Sampled at a 1-hour frequency. It contains the target OT and six distinct power load features (HUFL, HULL, MUFL, MULL, LUFL, LULL).
 - **ETTh2:** Sampled at a 1-hour frequency from a second transformer, with the same target and load feature structure settings.
 - **ETTm1:** Sampled at a 15-minute frequency, containing the same set of load features but with higher temporal resolution, capturing more granular fluctuations.
 - **ETTm2:** Sampled at a 15-minute frequency from a second transformer, matching the ETTm1 feature structure configuration.
- **Wind:** The target variable is stored as target, and the contextual variables include predicted humidity, pressure, temperature, wind direction, wind speed, and observed wind speed. The data is sampled at 15-minute intervals.

Short-term Electricity Price Forecasting

We utilize five widely adopted datasets from the electricity price forecasting benchmark established by Lago et al. (Lago et al. 2021). These datasets represent different regional day-ahead markets, each with a sampling frequency of 1 hour. The task is to forecast day-ahead prices (24 steps) using historical observations and exogenous variables.

- **BE (Belgium):** Represents the Belgian electricity market, including hourly electricity prices supplemented with national load forecasts and generation forecasts from the neighboring French grid.
- **DE (Germany):** Represents the German electricity market, recording hourly electricity prices with exogenous forecasts of zonal load in the Amprion TSO area, as well as wind and solar power generation.
- **FR (France):** Represents the French electricity market, comprising hourly electricity prices, grid load forecasts, and domestic generation forecasts.
- **NP (Nord Pool):** Represents the Nordic electricity market, containing hourly electricity prices along with exogenous forecasts of grid load and wind power generation.

- **PJM (Pennsylvania-New Jersey-Maryland):** Sourced from the PJM Interconnection in the United States, this dataset contains zonal electricity prices for the Commonwealth Edison (COMED) region, together with system-wide load forecasts and zonal load forecasts.

Detailed Baseline Descriptions

We evaluate CastFSR against a diverse set of representative baselines, ranging from classical statistical methods to state-of-the-art foundation models.

Statistical Methods.

- **ARIMA** (Hyndman and Khandakar 2008): A classic statistical method that models temporal patterns using autoregression, differencing, and moving averages to capture linear dependencies.
- **Prophet** (Taylor and Letham 2018): An additive regression model designed for business time series, which effectively decomposes data into trends, seasonality, and holiday effects.

Deep Learning Baselines.

- **DLinear** (Zeng et al. 2023): A simple yet effective MLP-based model that utilizes a decomposition layer to handle trend and seasonal components separately.
- **ConvTimeNet** (Cheng et al. 2025b): A deep hierarchical fully convolutional network that captures multi-scale temporal patterns through adaptive segmentation and deformable patching.
- **PatchTST** (Nie et al. 2023): A Transformer-based model that introduces channel independence and patch-based tokenization to capture local semantic information and reduce computational complexity.
- **iTransformer** (Liu et al. 2024): An inverted Transformer architecture that embeds the whole time series of each variate as a token and applies attention mechanisms across multivariate channels.
- **TimeXer** (Wang et al. 2024): An advanced Transformer framework designed to empower time series forecasting by incorporating and aligning exogenous variables.

Foundation Models.

- **TimesFM** (Das et al. 2024): A decoder-only foundation model developed by Google, pretrained on a massive corpus of over 100 billion real-world and synthetic time points. It utilizes a patch-based architecture to capture long-range temporal dependencies and enables accurate zero-shot forecasting across diverse domains.
- **Sundial** (Liu et al. 2025b): A family of pretrained time series foundation models designed to provide strong zero-shot and transfer forecasting performance.

LLM-based Methods.

- **OFA (One Fits All)** (Zhou et al. 2023): A generalized framework that leverages frozen pre-trained language models (e.g., GPT-2) for time series analysis. It adapts the LLM to forecasting tasks by fine-tuning only specific

Table 7: Summary of real-world time series datasets used in the experiments. The table details the domain, sampling frequency, variable dimensions, forecasting setting, and specific content of each dataset.

Setting	Dataset	Domain	Length	Variables	Frequency	Description
Long-term	ETTh1	Electricity	17,420	7	1 hour	Transformer oil temperature and 6 power load features.
Long-term	ETTh2	Electricity	17,420	7	1 hour	Transformer oil temperature and 6 power load features.
Long-term	ETTm1	Electricity	69,680	7	15 mins	Transformer oil temperature and 6 power load features.
Long-term	ETTm2	Electricity	69,680	7	15 mins	Transformer oil temperature and 6 power load features.
Long-term	Wind	Energy	48,673	7	15 mins	Target with meteorological forecasts and observed wind speed.
Short-term	BE	Energy	14,496	3	1 hour	Belgian market prices with load and generation forecasts.
Short-term	DE	Energy	14,496	3	1 hour	German market prices with load, wind, and solar forecasts.
Short-term	FR	Energy	14,496	3	1 hour	French market prices with load and generation forecasts.
Short-term	NP	Energy	14,496	3	1 hour	Nord Pool market prices with load and wind forecasts.
Short-term	PJM	Energy	14,496	3	1 hour	PJM (ComEd) prices with system and zonal load forecasts.

layers (such as positional embeddings and normalization layers) while keeping the self-attention and feedforward networks frozen.

- **Time-LLM** (Jin et al. 2024): A comprehensive framework that aligns time series modalities with the text space of LLMs using reprogramming techniques and prompt-as-prefix strategies.
- **TokenCast** (Tao et al. 2025): A value-tokenization method that converts numerical time series into language-compatible token sequences for LLM-based forecasting.
- **S²IP-LLM** (Pan et al. 2024): An LLM-based forecasting method that adapts pretrained language models to time series through structured semantic and instance-level prompting.
- **TimeReasoner** (Cheng et al. 2026): An approach that leverages the reasoning capabilities of LLMs to infer temporal dynamics and causal relationships within the time series data.
- **PromptCast** (Xue and Salim 2023): A prompt-based forecasting method that casts time series prediction into a language-model prompting task.

Agentic Forecasting Systems.

- **TimeSeriesScientist** (Zhao et al. 2025): An LLM-driven scientific agent for time series analysis and forecasting.
- **AlphaCast** (Zhang et al. 2025): A human-LLM co-reasoning framework for interactive TSF.

Detailed Implementation Settings

In this section, we provide the comprehensive configuration details for reproducing our experiments, including the backbone model specifications, training hyperparameters for both the Supervised Fine-Tuning (SFT) and Reinforcement Learning (RL) stages, and the setup for baseline comparisons.

CastFSR-Zero

To evaluate the effectiveness of the proposed Fast-Slow-Reflect workflow without task-specific optimization, we instantiate CastFSR in a training-free setting, denoted as CastFSR-Zero. Rather than directly generating numerical forecasts with the LLM, CastFSR-Zero treats the LLM as an agentic coordinator that orchestrates forecasting experts,

analytical tools, contextual evidence, and reflective evaluation. Specifically, we employ DeepSeek-V4-Flash as the coordinator, which is responsible for planning, tool invocation, contextual reasoning, and forecast refinement. During inference, we set the maximum output length to 32,768 tokens. Given historical observations and task descriptions, the coordinator first performs fast-thinking forecasting by profiling temporal characteristics and selecting appropriate forecasting experts from a model pool, including ARIMA, DLinear, PatchTST, iTransformer, and Chronos-2. These experts generate data-driven forecast priors, providing a numerical basis for subsequent contextual reasoning. Since CastFSR-Zero requires no parameter updates or task-specific training, it directly evaluates whether the proposed agentic workflow can improve context-aware forecasting.

CastFSR-R1

To investigate whether the proposed reasoning workflow can be learned by a compact forecasting agent, we further develop CastFSR-R1 through SFT and RL. Specifically, we adopt Qwen3-4B as the backbone model and distill high-quality reasoning trajectories generated by CastFSR-Zero across diverse forecasting tasks. The SFT stage is conducted with a learning rate of 2.0×10^{-6} to teach the model the fundamental Fast-Slow-Reflect workflow, including expert selection, contextual reasoning, and reflective validation. We then apply the GRPO algorithm for reinforcement learning with a learning rate of 2.0×10^{-7} to further optimize the agent’s decision-making policy. The RL stage uses a global batch size of 128, a group size of $G = 5$, and maximum prompt and response lengths of 20,480 and 6,144 tokens, respectively. The model is trained on 16 Ascend NPUs with distilled cross-domain trajectories to improve generalization across diverse forecasting scenarios. Unlike CastFSR-Zero, which relies on an external LLM coordinator, CastFSR-R1 integrates the Fast-Slow-Reflect reasoning capability into a compact model, enabling efficient deployment while maintaining the ability to coordinate numerical forecasting, contextual reasoning, and knowledge-constrained reflection.

Prompt Construction Details

As shown in Template 1, CastFSR constructs prompts according to the three-stage Fast-Slow-Reflect workflow. Each

LLM Coordinator	ETTh1		ETTm1		Wind		DE		FR		NP	
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
GPT-5.6-sol	<u>0.081</u>	0.213	<u>0.055</u>	0.168	1.166	0.741	0.400	0.414	4.056	0.505	0.218	<u>0.266</u>
LongCat-2.0	0.089	0.219	0.056	0.168	1.739	0.939	0.383	0.412	<u>3.943</u>	0.507	0.221	0.271
DeepSeek V4 Pro	<u>0.081</u>	<u>0.212</u>	0.054	0.165	<u>1.314</u>	0.810	0.403	0.417	4.060	0.505	<u>0.216</u>	0.267
DeepSeek V4 Flash	<u>0.081</u>	<u>0.212</u>	<u>0.055</u>	0.168	1.596	0.886	<u>0.393</u>	0.408	3.939	<u>0.501</u>	0.221	0.277
GLM-5.2	0.080	0.210	<u>0.055</u>	<u>0.167</u>	1.332	<u>0.806</u>	0.405	0.427	3.966	0.498	0.207	0.260

Table 8: Full training-free forecasting performance of CastFSR-Zero with different pretrained LLM coordinators on all datasets reported for this analysis. Lower values indicate better performance.

Variant	ETTh1		ETTh2		ETTh1		ETTh2		Wind		BE		DE		FR		NP		PJM	
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
w/o SFT	<u>0.081</u>	<u>0.213</u>	0.274	0.367	0.055	0.168	0.141	0.254	<u>1.795</u>	<u>0.920</u>	<u>0.959</u>	<u>0.384</u>	0.435	0.436	4.059	0.504	0.237	0.280	<u>0.144</u>	<u>0.267</u>
w/o RL	0.082	0.215	<u>0.268</u>	<u>0.364</u>	0.055	<u>0.170</u>	<u>0.139</u>	<u>0.253</u>	1.813	<u>0.920</u>	0.961	0.395	<u>0.392</u>	<u>0.409</u>	<u>4.047</u>	<u>0.503</u>	<u>0.217</u>	<u>0.268</u>	0.149	0.275
CastFSR-R1	0.077	0.210	0.242	0.351	0.055	0.168	0.127	0.237	1.757	0.918	0.927	0.375	0.386	0.405	4.032	0.502	0.172	0.247	0.133	0.260

Table 9: Full performance comparison of CastFSR-R1 with variants removing supervised fine-tuning (w/o SFT) or reinforcement learning (w/o RL). Lower values indicate better performance.

prompt contains the shared forecasting situation, including task metadata, domain priors, timestamp scope, sampling frequency, target variable, and available future-known covariates. The remaining prompt content and admissible actions are then adjusted to match the current stage.

- **Stage 1: Fast-thinking Forecasting.** The prompt provides the historical look-back window and task context, and asks the agent to profile temporal patterns, select diagnostic evidence, and route the instance to a suitable lightweight forecaster. The output of this stage is a data-driven forecast prior together with structured evidence about trend, seasonality, local dynamics, data quality, or residual behavior.
- **Stage 2: Slow Deliberative Reasoning.** The prompt includes the forecast prior from Stage 1, the retained diagnostic evidence, recent historical observations, and available contextual variables. It asks the agent to determine whether contextual signals are relevant to the forecast horizon, infer their direction and temporal scope, and refine the numerical prior without replacing it blindly.
- **Stage 3: Reflective Evaluation.** The prompt provides the candidate forecast, the forecast prior, the contextual reasoning trace, and task-specific constraints. It asks the agent to check temporal consistency, contextual support, domain validity, timestamp format, and numerical feasibility, then apply localized corrections when necessary before producing the final forecast sequence.

Additional Experimental Results

Table 8 reports the full training-free LLM coordinator comparison corresponding to the representative results in the main text. Table 9 reports the full SFT/RL ablation results across all benchmark datasets.

Effect of Different LLM Coordinators. Table 8 evaluates the training-free CastFSR-Zero with different pretrained

LLM coordinators to investigate whether the effectiveness of our agentic workflow depends on a specific foundation model. Overall, CastFSR-Zero achieves consistent performance, demonstrating the generality of the Fast–Slow–Reflect workflow. Although different coordinators exhibit complementary strengths across datasets, no single LLM consistently dominates all forecasting scenarios. For example, GLM-5.2 achieves the best performance on ETTh1 and NP, while DeepSeek V4 Pro performs best on ETTh1. DeepSeek V4 Flash obtains competitive results on multiple datasets, achieving the lowest MAE on DE and the lowest MSE on FR, while GPT-5.6-sol achieves the strongest results on Wind. These variations indicate that forecasting performance depends not only on the intrinsic reasoning ability of the LLM, but also on its ability to coordinate forecasting experts, interpret contextual evidence, and perform reflective validation. By decoupling numerical forecasting from high-level reasoning, CastFSR-Zero enables different LLMs to leverage specialized forecasting experts and structured analytical tools. The results verify that the Fast–Slow–Reflect workflow provides a model-agnostic framework for context-aware forecasting.

Effect of Training Strategies. Table 9 investigates the contribution of supervised fine-tuning (SFT) and reinforcement learning (RL) in CastFSR-R1 by removing each training stage individually. Removing either SFT or RL generally degrades performance across benchmarks, with occasional ties on individual metrics, demonstrating that both stages play complementary roles in learning effective forecasting agents. Specifically, the variant without SFT suffers from notable performance drops on most datasets, indicating that supervised fine-tuning provides essential guidance for learning the basic Fast–Slow–Reflect workflow, including expert coordination, contextual reasoning, and reflective validation. Without this initialization, the model struggles to effectively organize multi-stage reasoning and tool uti-

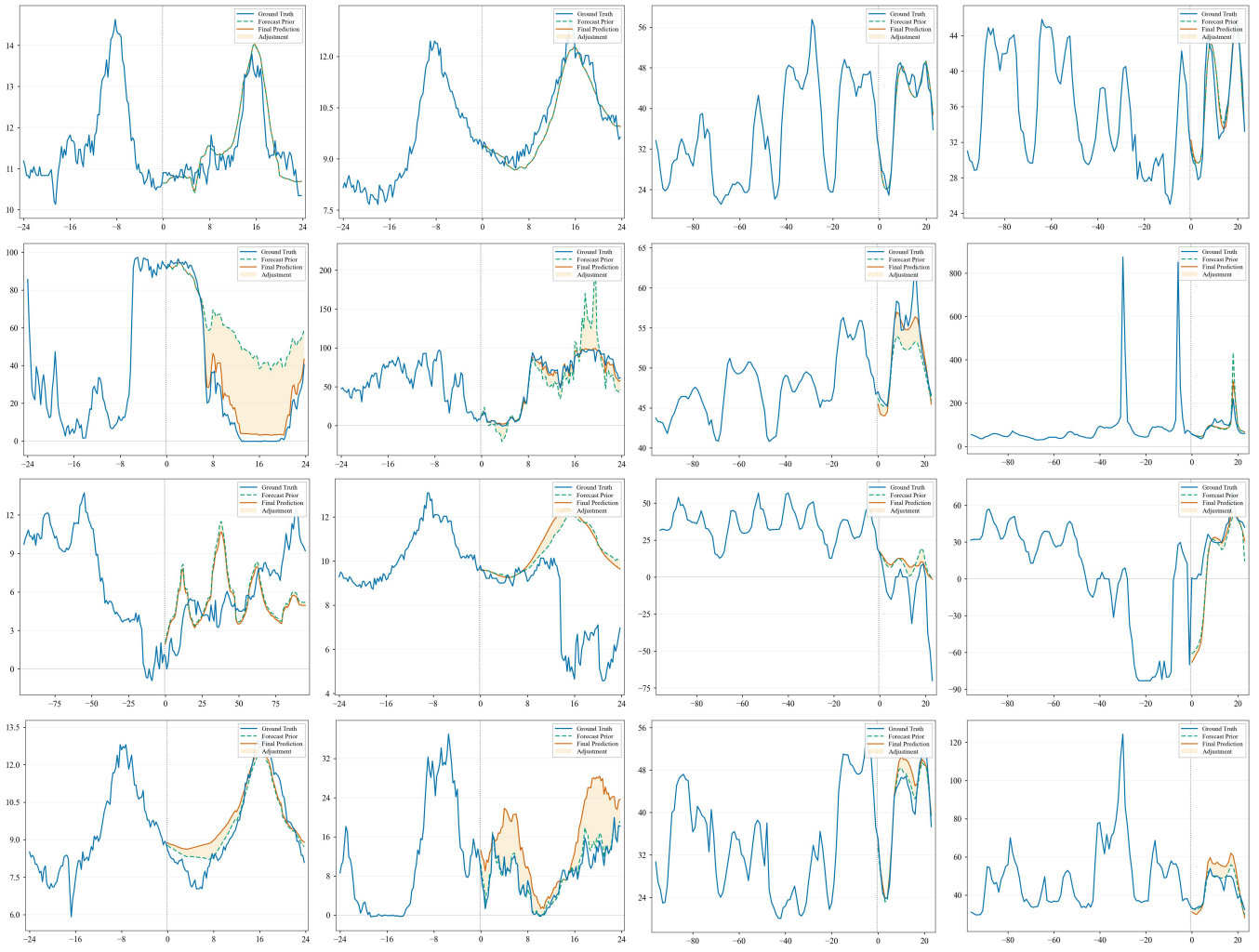


Figure 7: Case study of adaptive look-back window selection in CastFSR. The upper examples show successful cases where the retrieved contextual windows provide relevant historical evidence for future dynamics, enabling effective refinement of the forecast prior. The lower examples show challenging cases where inappropriate context windows provide limited or misleading evidence, resulting in insufficient or excessive adjustments.

lization. Meanwhile, removing RL also generally leads to degradation, especially on several forecasting benchmarks, suggesting that reinforcement learning further improves the agent’s decision-making ability by optimizing workflow execution and forecast refinement beyond supervised imitation. Overall, the complete CastFSR-R1 with both SFT and RL achieves the best or tied-best performance, verifying that supervised learning establishes the reasoning foundation while reinforcement learning enhances adaptive decision-making.

Visualization Analysis

Figure 7 illustrates how the selected look-back windows influence the refinement process of CastFSR. The upper examples represent successful cases where the retrieved historical contexts exhibit strong relevance to future dynamics. By identifying consistent patterns or regime transitions from the selected windows, CastFSR appropriately adjusts the forecast

prior and produces predictions closer to the ground truth. In contrast, the lower examples correspond to challenging scenarios where the selected contexts provide insufficient or misleading evidence. Consequently, the contextual reasoning process may introduce inaccurate adjustments or fail to fully correct the initial forecast. These observations highlight that context-aware forecasting is not simply a matter of incorporating more historical information, but requires adaptively identifying informative temporal windows that match the underlying dynamics. The case study verifies that the Fast-Slow-Reflect workflow can leverage context-specific evidence to refine numerical priors while avoiding unnecessary modifications.

Fast–Slow Agentic Forecasting Prompt

Role & Context: Act as the forecasting policy of a context-aware time-series agent. All decisions remain **target-blind**, grounded in historical observations, task and domain priors, temporal context, available future-known covariates, diagnostic evidence, and auxiliary forecasts.

Stage 1—Fast-thinking Forecasting: Construct a pattern-based numerical prior through evidence acquisition and context-aware predictor routing.

Evidence Acquisition. Follow a “Stop-and-Look” policy and select the diagnostic view that best resolves the dominant uncertainty:

- `extract_basic_statistics`: global structure and dependence;
- `extract_within_channel_dynamics`: local dynamics and regime shifts;
- `extract_forecast_residuals`: forecast bias and residual structure;
- `extract_data_quality`: corruption, saturation, and dropout;
- `extract_event_summary`: segment-level temporal events.

Exactly one diagnostic action is admitted before prediction.

Context-Aware Predictor Routing. Route the task to a single auxiliary predictor according to the situation context and acquired evidence:

- **ARIMA**: linear dynamics with stable seasonality;
- **DLinear**: decomposable trend and seasonal dynamics;
- **PatchTST**: regular dependence and multi-scale seasonality;
- **iTransformer**: pronounced cross-channel dependence;
- **Chronos2**: irregular, nonlinear, or shifting regimes.

The preferred model acts only as a tie-breaker. Coarse target-blind calibration is reserved for evidence-supported global bias.

Stage 2—Slow Deliberative Reasoning: Treat the auxiliary trajectory as a numerical baseline rather than a final answer, preserve its temporal alignment, and reason through two coupled views:

- **Situation Synthesis**: infer correction direction, reliability, lag, and temporal scope from regime, domain, and exogenous evidence;
- **Forecasting Reasoning**: reconcile this guidance with historical features and the auxiliary level, shape, and phase.

Refinement follows natural cycles and change points. Its scope and magnitude are regularized by evidence strength, enabling conservative local adjustment under uncertainty and larger yet localized correction under coherent evidence.

Stage 3—Reflective Evaluation

Consistency Audit: Synthesize rule diagnostics with the situation context, feature evidence, auxiliary trajectory, candidate forecast, and prior reflection trace. A target-blind, dataset-specific rule gate activates reflection only for a critical violation or severe conflicts spanning multiple rule families.

Adaptive Re-entry:

- **Evidence-level inconsistency**: return to Stage 1 and reconsider feature, regime, exogenous evidence, and predictor routing;
- **Reasoning-level inconsistency**: retain valid evidence and the auxiliary trajectory, and revisit Stage 2.

The re-entry depth is matched to the diagnosed source of error, preserving valid intermediate state while revisiting only the uncertain reasoning path.

Structured Output: The rationale links situation evidence, auxiliary diagnostics, consistency validation, and uncertainty-aware decision making.

```
<think>
Part I - Situation synthesis and analysis
Part II - Situation-guided forecasting reasoning
</think>
<answer>
YYYY-MM-DD HH:MM:SS finite_decimal
...
</answer>
```